

MagMat-Discovery

Physics-Informed Machine Learning and Screening for Magnetic Materials

System Architecture, Methodology & Model Validation

Technical Report & Model Guide

Author: Sachin Poudel

Project: MagMat-Discovery Platform

Web Platform: Interactive Streamlit Discovery Engine

Date: September 2026

Contents

Executive Summary	1
Glossary of Key Terms	2
1 Data and Partitioning	3
1.1 Data Sources	3
1.2 Data Cleaning and Grouped Splits	3
2 Methodology	4
2.1 Architecture and Provenance	4
2.2 Representations and Structure Synthesis	4
2.3 Ensembles, Cascades, and Uncertainty	5
3 Model Validation	7
3.1 Benchmark Performance	7
3.2 Parity Analysis	7
3.3 Manifolds, Diagnostics, and Screening	8
4 Interactive Platform	11
5 Summary and Outlook	11
A Physical Feature Formulations	13
A.1 Goodenough-Kanamori Superexchange	13
A.2 Dzyaloshinskii-Moriya Interaction (DMI) Proxy	13
A.3 Crystal Field and Stevens Operator Hamiltonian	13
B MagFormer Architecture Details	14
B.1 Node Initialization and Projection	14
B.2 Pair-Bias Self-Attention	14
B.3 Leakage-Free Composition Pooling	14

Executive Summary

Discovering permanent magnets with reduced rare-earth content is important for electric vehicles, wind turbines, and industrial machinery. Useful candidates must meet three criteria: Curie temperature $T_C > 400$ K, saturation magnetization $\mu_0 M_s > 1.0$ T, and coercivity $H_c > 800$ kA/m. However, machine learning for magnetic materials is challenging because intrinsic properties depend on atomic chemistry while extrinsic properties depend on grain microstructure. Furthermore, random data splits cause leakage between identical compositions, and unconstrained models can make unphysical predictions.

MagMat-Discovery addresses these challenges through four objectives:

1. **Multi-Target Modeling:** Predict 8 intrinsic and extrinsic magnetic properties covering phase, transition temperatures, and hysteresis behavior.
2. **Grouped Evaluation:** Evaluate models with 5-fold grouped cross-validation by chemical formula to prevent data leakage.
3. **Physics Cascades:** Connect intrinsic predictions (K_1, M_s) to extrinsic properties (H_c and $(BH)_{\max}$) using micromagnetic equations and physical bounds.
4. **Calibrated Uncertainty:** Provide 90% conformal intervals for every continuous property alongside an interactive screening app.

By combining experimental data, physical features, the **MagFormer** graph model, ensembles, and conformal calibration, the platform accurately predicts magnetic properties and provides physically bounded rankings for magnet discovery.

Keywords: Magnetic Materials, Physics-Informed Machine Learning, Graph Transformers, Conformal Calibration, Coercivity, Permanent Magnets, Materials Screening.

Glossary of Key Terms

Table 1 summarizes the primary magnetic properties and machine learning terminology used throughout this report.

Table 1: Glossary of magnetic properties, physical units, and computational terms.

Term	Unit	Definition
T_C (Curie Temp.)	K	Temperature above which a ferromagnet loses spontaneous magnetization.
T_N (Néel Temp.)	K	Temperature above which an antiferromagnet becomes paramagnetic.
M_s (Saturation Mag.)	emu/g or T	Maximum magnetization in an applied field, reported in emu/g or $\mu_0 M_s$ (T).
H_c (Coercivity)	A/m or kA/m	Magnetic field required to reduce magnetization to zero after saturation.
K_1 (Anisotropy Constant)	J/m ³	Energy density required to rotate magnetization away from the preferred crystal axis.
$(BH)_{\max}$ (Energy Product)	kJ/m ³	Maximum stored magnetic energy density; main figure of merit for permanent magnets.
θ_p (Paramagnetic Curie Temp.)	K	High-temperature intercept from Curie-Weiss susceptibility fitting.
FM / AFM / NM	Discrete	Ferromagnetic, Antiferromagnetic, or Non-magnetic ground-state ordering.
Out-of-Fold (OOF)	–	Predictions on held-out validation data, avoiding data leakage.
Conformal Range	Physical	Uncertainty window covering the true value 90% of the time.
dex	log ₁₀	Decimal order of magnitude. An error of 0.65 dex means a factor of $10^{0.65} \approx 4.5$.

1 Data and Partitioning

1.1 Data Sources

The platform uses three primary data sources:

- **NEMAD Database:** 51,394 experimental records [1] across all 8 target properties.
- **Materials Project:** Crystal symmetries (space groups 1–230, crystal systems, and densities) [2]. DFT energies and moments are excluded to prevent target leakage.
- **Bilbao MAGNDATA:** Magnetic space groups and Shubnikov symmetry classifications.

1.2 Data Cleaning and Grouped Splits

Chemical formulas are simplified to lowest integer ratios with `pymatgen` (e.g., $\text{Fe}_6\text{Pt}_2 \rightarrow \text{Fe}_3\text{Pt}$). Properties spanning multiple orders of magnitude (H_c , K_1 , $(BH)_{\max}$) are log-transformed.

To evaluate generalization to unseen materials, models are tested with 5-fold cross-validation grouped by chemical formula (`GroupKFold`). All entries with the same formula or crystal phase stay in the same split, ensuring test folds contain only new compounds.

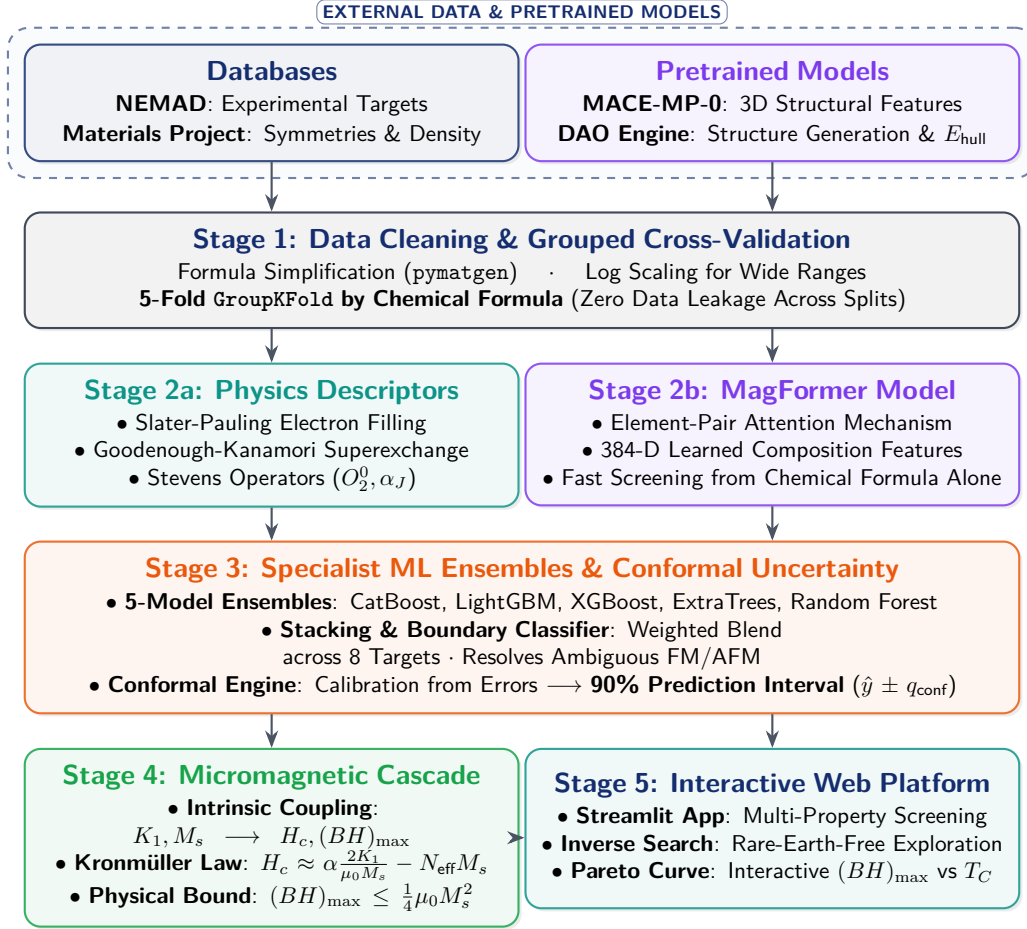


Figure 1: Workflow of MagMat-Discovery, showing external resources, in-house representations, 5-model ensembles, physical cascades, and candidate screening.

2 Methodology

2.1 Architecture and Provenance

Figure 1 shows the overall workflow of MagMat-Discovery, distinguishing external resources from in-house algorithms (Table 2). External databases and foundation models provide experimental records, crystal symmetries, and 3D crystal generation. All regressors, phase classifiers, micromagnetic cascades, and uncertainty calibrators were trained and evaluated in-house.

Table 2 lists the data sources, pretrained models, and in-house components used in the pipeline.

2.2 Representations and Structure Synthesis

The pipeline represents materials using three complementary methods:

1. Physics Descriptors: Features organized into four tiers (Appendix A):

- Tier 1 (Composition):** Mass fractions, atomic numbers, covalent radii, and electronegativities.
- Tier 2 (Electronic Structure):** Valence electron count relative to the Slater-Pauling peak ($\text{VEC} \approx 8.3$), exchange rules, and spin-orbit coupling ($\propto Z^4$).
- Tier 3 (Crystal Field):** Crystal family indicators and Stevens factors (O_2^0, α_J) for easy-axis preference.

Table 2: Data sources, models, and algorithmic components in MagMat-Discovery.

Component	Type	Source	Role in Pipeline
Materials Project [2]	External DB	Open DFT repository	Crystal symmetries and densities (energies/moments excluded).
NEMAD Database [1]	External DB	Experimental literature	Target measurements for all 8 magnetic properties.
MACE-MP-0 [3]	Pretrained	Equivariant GNN	3D structural features for compounds with known coordinates.
DAO Engine [4]	Pretrained	Diffusion model	Generates 3D unit cells (DAO-G) and checks stability E_{hull} (DAO-P).
MagFormer	In-House	Trained in this work	Maps chemical formulas to 384-D features without 3D coordinates.
Physics Descriptors	In-House	Domain physics	Electron counts (VEC), superexchange proxies, Stevens factors.
5-Model Ensembles	In-House	5-fold grouped training	Blends CatBoost, LightGBM, XGBoost, ExtraTrees, Random Forest.
Boundary Classifier	In-House	Ambiguous data	Resolves ambiguous FM vs. AFM phase predictions.
Micromagnetic Cascade	In-House	Micromagnetic theory [5]	Enforces Kronmüller coercivity and $(BH)_{\text{max}} \leq \frac{1}{4}\mu_0 M_s^2$.
Conformal Engine	In-House	Validation residuals [6]	Provides 90% prediction intervals for continuous properties.

- *Tier 4 (3D Features)*: Atomic representations from MACE-MP-0 [3] when 3D coordinates exist.

2. MagFormer Model: For screening materials without crystal coordinates, **MagFormer** maps chemical formulas to 384-dimensional feature vectors using self-attention (Appendix B).

3. DAO Structure Generation: When 3D coordinates are needed, the pretrained DAO model [4] generates unit cells (DAO-G) and calculates stability (E_{hull} via DAO-P). Materials with $E_{\text{hull}} < 0.05$ eV/atom are considered synthesizable.

2.3 Ensembles, Cascades, and Uncertainty

We predict each property using an ensemble of five machine learning models: CatBoost, LightGBM, XGBoost, ExtraTrees, and Random Forest. Their predictions are combined through weighted averaging (non-negative least squares). For magnetic phase, a secondary classifier resolves borderline cases where ferromagnetic and antiferromagnetic probabilities are close ($|P(\text{FM}) - P(\text{AFM})| < 0.20$).

To prevent predictions that violate the laws of physics, the pipeline uses a step-by-step physical cascade:

1. Intrinsic properties (K_1 and M_s) are predicted first.
2. These predictions feed into the coercivity (H_c) and energy product ($(BH)_{\text{max}}$) models,

ensuring coercivity follows the Kronmüller relationship [5]:

$$H_c \approx \alpha_{\text{micro}} \frac{2K_1}{\mu_0 M_s} - N_{\text{eff}} M_s \quad (1)$$

where α_{micro} accounts for grain defects and N_{eff} is the demagnetizing factor.

3. Predicted energy products are capped at the theoretical limit: $(BH)_{\text{max}} \leq \frac{1}{4} \mu_0 M_s^2$.

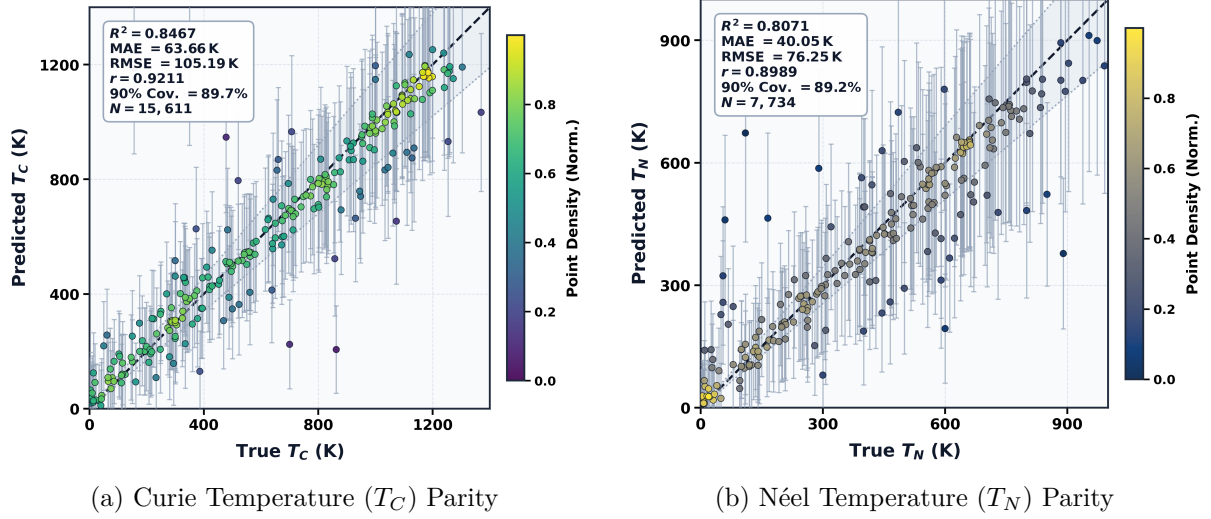
Every continuous prediction includes a calibrated uncertainty range using conformal prediction [6]. By taking the 90th percentile of absolute errors on held-out validation data:

$$q_{\text{conf}} = \text{Quantile}_{0.90}(\{|y_i - \hat{y}_i|\}_{i=1}^n) \quad (2)$$

we provide an error window $[\hat{y} - q_{\text{conf}}, \hat{y} + q_{\text{conf}}]$ that covers the true value 90% of the time.

Table 3: Comprehensive 5-fold GroupKFold cross-validation results across all 8 magnetic targets.

#	Target Property	Unit	Scale	Count n	Primary Metric	Mean Error	Practical Use
1	Ground-State Phase	Discrete	Class	35,045	91.9% Accuracy	Macro F_1 : 0.908	Strong (Reliable phase filtering)
2	Curie Temperature (T_C)	K	Linear	15,518	$R^2 = 0.816$	MAE = 69.8 K	Strong (Thermal threshold filter)
3	Néel Temperature (T_N)	K	Linear	7,734	$R^2 = 0.741$	MAE = 48.6 K	Strong (Accurate AFM boundary)
4	Anisotropy Constant (K_1)	J/m ³	Symlog	3,938	$R^2 = 0.461$	MAE = 0.95 dex	Moderate (Easy-axis candidate ranking)
5	Coercivity (H_c)	A/m	log ₁₀	6,980	$R^2 = 0.622$	MAE = 0.65 dex	Moderate (Hard vs soft screening)
6	Saturation Magnetization (M_s)	emu/g	Linear	7,619	$R^2 = 0.539$	MAE = 26.1 emu/g	Moderate (Moment estimation)
7	Energy Product ($(BH)_{\max}$)	kJ/m ³	log ₁₀	1,379	$R^2 = 0.507$	MAE = 0.27 dex	Moderate (Bounded density estimate)
8	Paramagnetic Curie Temp. (θ_p)	K	Linear	5,474	$R^2 = 0.607$	MAE = 64.7 K	Moderate (Exchange coupling sign)

Figure 2: Parity plots for Curie (T_C) and Néel (T_N) transition temperatures. Both models show strong accuracy across operating temperatures ($R^2 = 0.82$ and 0.74), with slight scatter below 100 K.

3 Model Validation

3.1 Benchmark Performance

Models are evaluated using 5-fold cross-validation grouped by chemical formula (Section 1.2). Performance is measured by the coefficient of determination (R^2) and mean absolute error (MAE). MAE is reported in physical units (K, emu/g) or decimal orders of magnitude (log₁₀ error) for wide-range properties (H_c , K_1 , $(BH)_{\max}$). Table 3 reports benchmark results across all 8 targets.

3.2 Parity Analysis

Performance reflects the difference between intrinsic properties (set by chemistry) and extrinsic properties (set by microstructure):

Transition Temperatures (T_C and T_N): Curie and Néel temperatures achieve high accuracy ($R^2 = 0.82$ and 0.74 , Figure 2) because ordering temperatures depend directly on chemical bonding and composition.

Magnetization and Coercivity (M_s and H_c): Saturation magnetization reaches $R^2 = 0.54$ (MAE = 26.1 emu/g, Figure 3). Coercivity spans four orders of magnitude (10^2 to 10^6 A/m). Because coercivity depends on grain boundaries and defects not captured by formula alone, predictions show wider scatter (MAE = 0.65 in log₁₀ scale). Nonetheless, the overall trend ($R^2 = 0.62$) reliably separates hard from soft magnets.

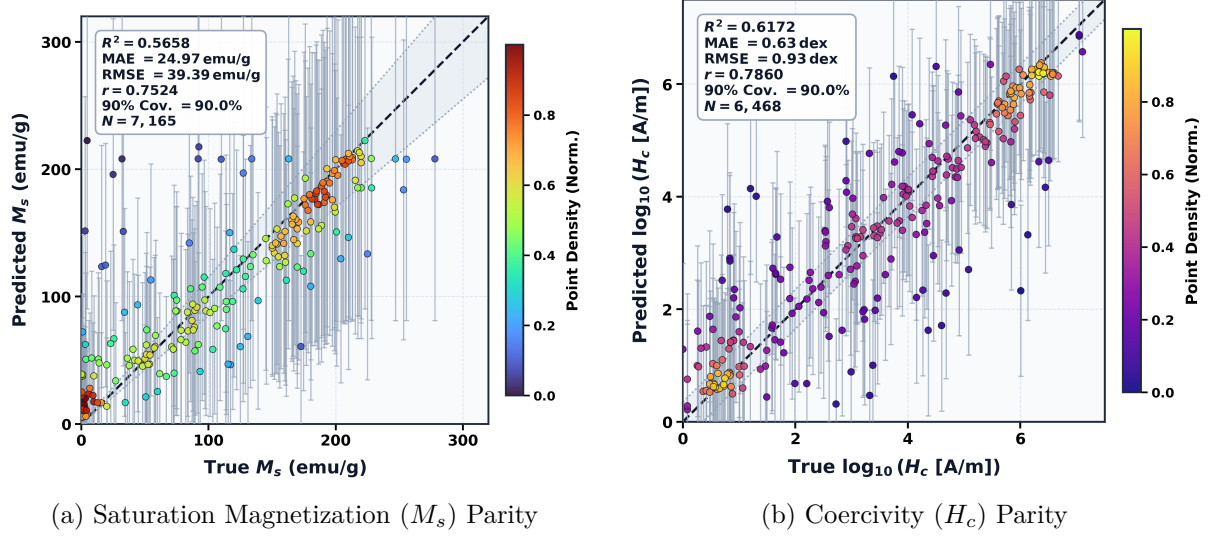


Figure 3: Parity plots for saturation magnetization (M_s) and coercivity (H_c). Magnetization shows moderate predictive utility ($R^2 = 0.54$). Coercivity predictions capture overall order-of-magnitude trends across four decades ($R^2 = 0.62$, MAE = 0.65 dex), serving as an effective pre-filter for hard magnetic behavior.

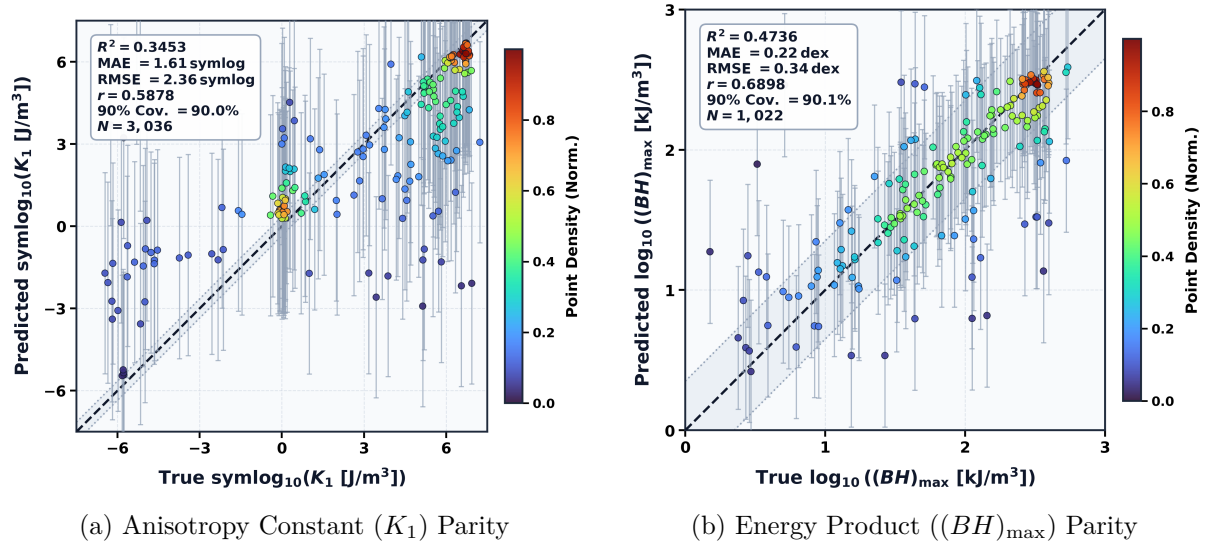


Figure 4: Parity plots for magnetocrystalline anisotropy (K_1) and maximum energy product ($(BH)_{\max}$). Cascade modeling ensures $(BH)_{\max}$ remains within thermodynamic limits. The moderate scores ($R^2 = 0.46$ and 0.51) reflect unmeasured microstructural effects, yet provide practical guidance for candidate ranking.

Anisotropy and Energy Product (K_1 and $(BH)_{\max}$): Anisotropy K_1 ($R^2 = 0.46$, Figure 4) identifies strong easy-axis candidates ($K_1 > 10^6$ J/m³). For energy product $(BH)_{\max}$ ($R^2 = 0.51$, MAE = 0.27 in $\log_{10}$ scale), the cascade enforces $(BH)_{\max} \leq \frac{1}{4}\mu_0 M_s^2$, preventing unphysical predictions.

3.3 Manifolds, Diagnostics, and Screening

To assess representation quality, classification discrimination, statistical uncertainty, and practical candidate discovery:

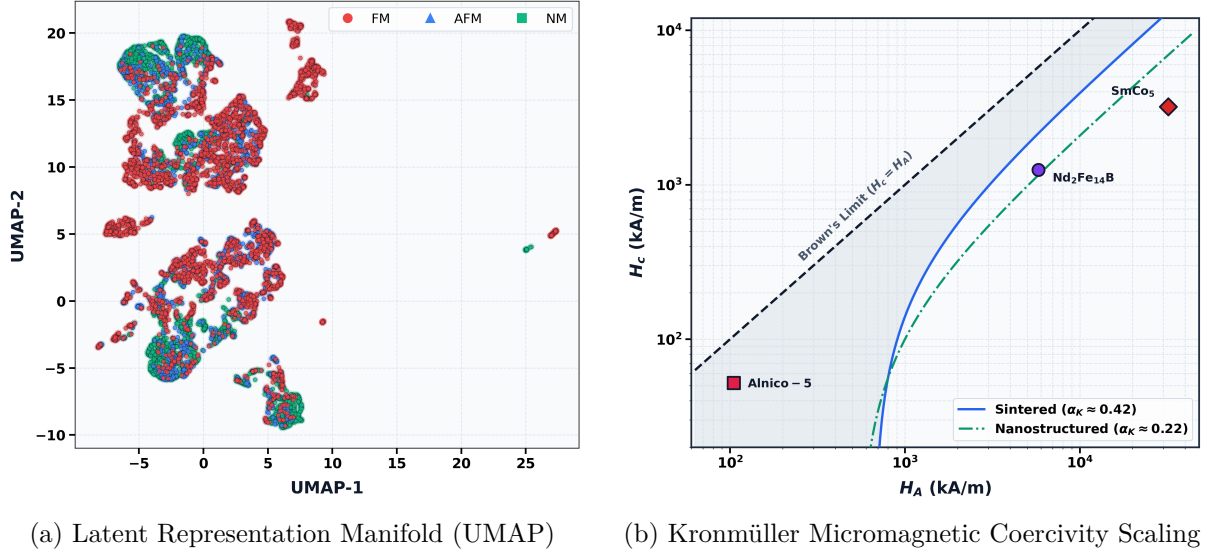


Figure 5: Latent representations and micromagnetic validation. (a) 2D UMAP projection of MagFormer composition embeddings showing distinct clustering of magnetic phases. (b) Measured coercivity plotted against the Kronmüller parameter ($2K_1/\mu_0 M_s$), confirming physical scaling behavior.

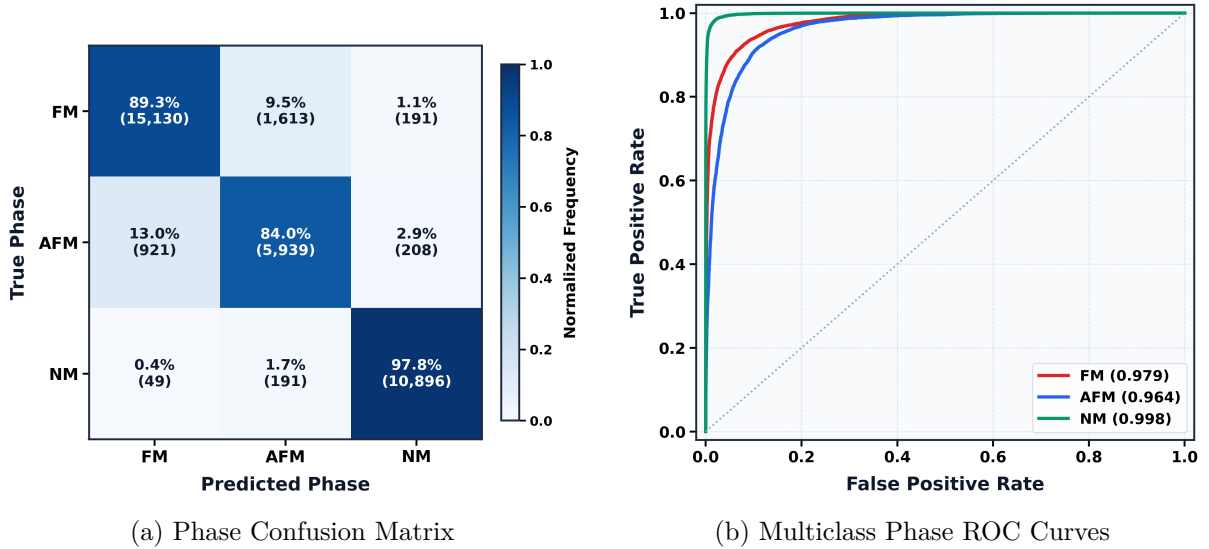


Figure 6: Ground-state magnetic phase classification diagnostics. (a) Confusion matrix showing balanced discrimination across phases. (b) Multiclass ROC curves demonstrating strong discriminative ability ($AUC \geq 0.96$) across Ferromagnetic, Antiferromagnetic, and Non-magnetic classes.

Learned Representations and Micromagnetic Scaling: Figure 5(a) maps MagFormer features via UMAP, showing distinct clusters for ferromagnetic, antiferromagnetic, and non-magnetic materials. Figure 5(b) confirms that coercivity scales with the Kronmüller parameter ($2K_1/\mu_0 M_s$).

Magnetic Phase Classification: Phase classification achieves 91.9% accuracy ($AUC \geq 0.96$, Figure 6), with the boundary classifier reaching $F_1 = 0.82$ on the smaller antiferromagnetic class.

Uncertainty Calibration: The conformal calibration curve (Figure 7(a)) closely tracks the ideal diagonal, confirming that 90% intervals achieve correct coverage. Residuals remain

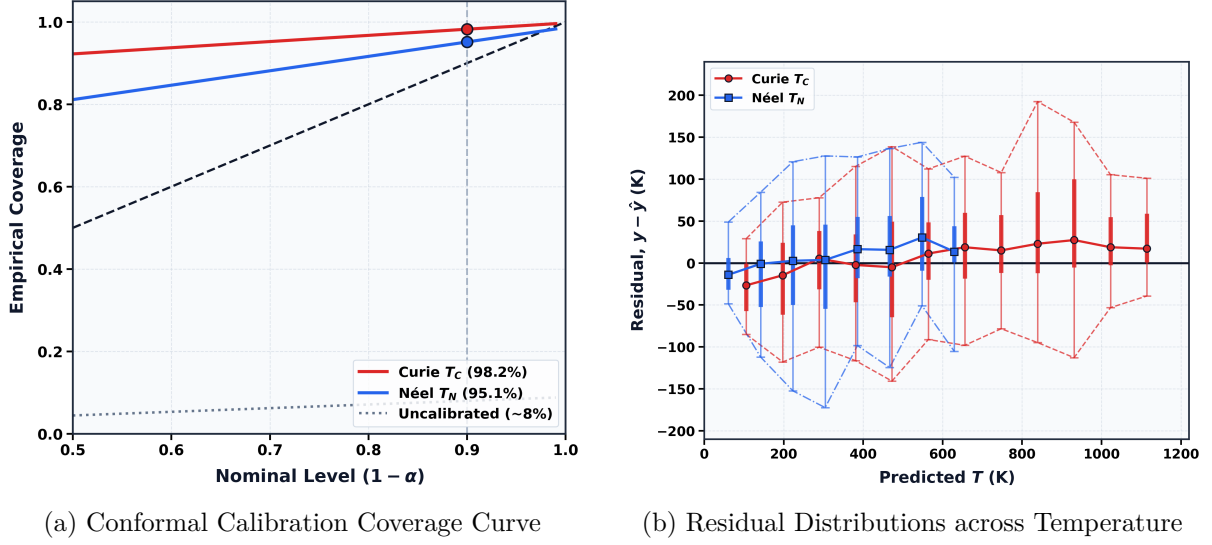


Figure 7: Uncertainty calibration and residual diagnostics. (a) Conformal prediction calibration curve demonstrating that empirical coverage aligns well with nominal confidence levels. (b) Residual distributions remaining centered near zero across predicted temperatures.

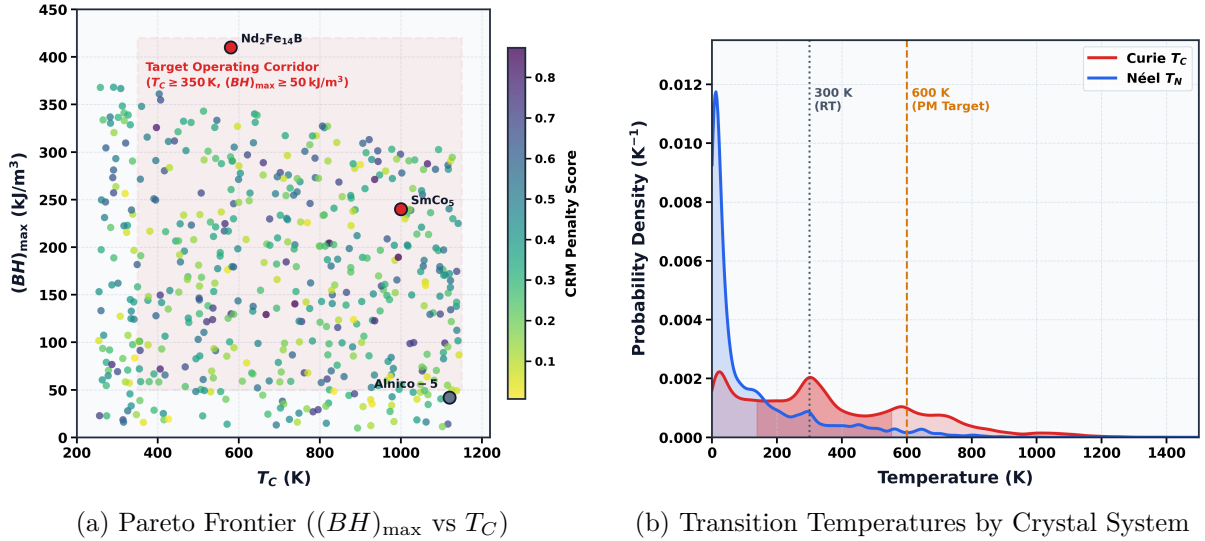


Figure 8: Candidate screening and symmetry profiles. (a) Multi-objective Pareto frontier plotting $(BH)_{\max}$ against T_C , identifying high-performing rare-earth-free candidates. (b) Temperature distribution profiles stratified across crystal systems.

centered near zero across operating temperatures (Figure 7(b)).

Candidate Screening: The Pareto frontier (Figure 8(a)) identifies candidates balancing $(BH)_{\max}$ and T_C . Crystal system distributions (Figure 8(b)) confirm that tetragonal and hexagonal systems provide the highest share of high- T_C , high-anisotropy materials.

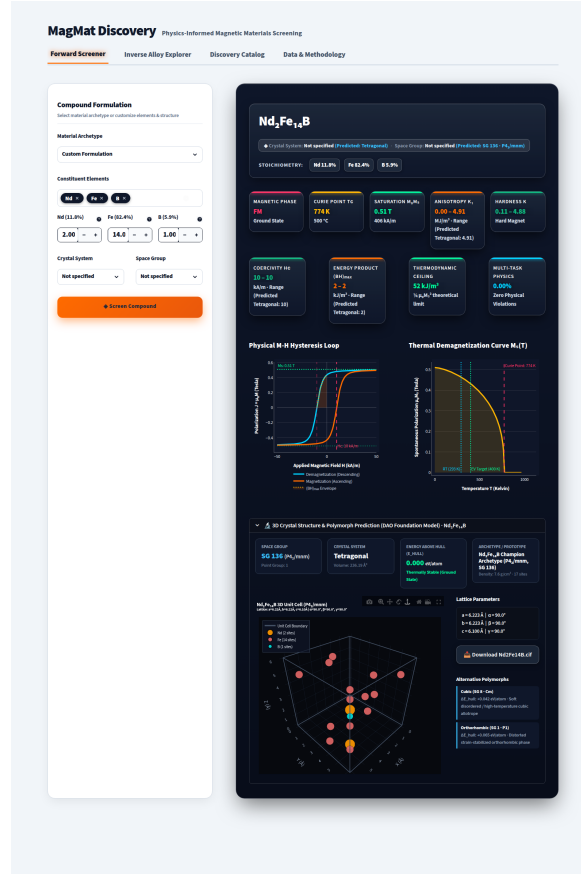


Figure 9: MagMat-Discovery interactive web platform interface. The forward screening dashboard displays predicted magnetic properties, multi-task physical consistency checks, simulated M - H hysteresis loop, thermal demagnetization curve $M_s(T)$, and interactive 3D crystal structure for $\text{Nd}_2\text{Fe}_{14}\text{B}$.

4 Interactive Platform

The MagMat-Discovery web application provides an interactive interface for candidate evaluation and screening (Figure 9), organized into four primary modules:

- **Forward Screener:** Users enter a formula or select reference magnets ($\text{Nd}_2\text{Fe}_{14}\text{B}$, SmCo_5 , Alnico, Fe_{16}N_2) to view property predictions with 90% intervals, simulated M - H loops, $M_s(T)$ curves, and 3D crystal structures.
- **Inverse Explorer:** Users set targets (e.g., $T_C \geq 500$ K, $(BH)_{\max} \geq 120$ kJ/m³) and filter by composition, such as excluding critical rare earths (Nd, Dy, Sm).
- **Discovery Catalog:** Browsing and export of material records, with filtering along the Pareto efficiency frontier.
- **Reference Tab:** Dataset statistics, space group distributions, and validation metrics displayed in the app.

5 Summary and Outlook

MagMat-Discovery combines physical principles, graph neural networks, and micromagnetic rules to screen permanent magnets:

- **Accurate Screening:** Magnetic phase classification (91.9% accuracy) and ordering temperatures ($R^2 = 0.82$ for T_C) serve as reliable initial filters.
- **Physical Ranking:** Extrinsic properties ($H_c, K_1, (BH)_{\max}$) achieve $R^2 = 0.46$ – 0.62 , providing a grounded pre-filter to prioritize candidates for synthesis.

Future work will incorporate grain-boundary phase models, domain wall pinning, and integration with automated synthesis platforms.

References

- [1] NEMAD Consortium, *A non-equilibrium magnetic materials database and machine learning framework for magnetic materials discovery*, Nature Communications, **16**, 2025, doi:10.1038/s41467-024-55500-7.
- [2] A. Jain et al., *Commentary: The Materials Project: A materials genome approach to accelerating materials innovation*, APL Materials, **1**(1), 011002, 2013.
- [3] I. Batatia, D. P. Kovacs, G. Simm, C. Ortner, and G. Csanyi, *MACE: Higher order equivariant and message passing neural networks for materials*, Advances in Neural Information Processing Systems (NeurIPS), **35**, 11423–11436, 2022.
- [4] Z. Huang et al., *Siamese Foundation Models for Crystal Structure Prediction*, Nature Communications, **17**, 2026, doi:10.1038/s41467-026-72362-3.
- [5] H. Kronmüller and M. Fähnle, *Micromagnetism and the Microstructure of High-Technology Materials*, Cambridge University Press, 2003.
- [6] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*, Springer, 2005.
- [7] J. M. D. Coey, *Magnetism and Magnetic Materials*, Cambridge University Press, 2010.

A Physical Feature Formulations

This appendix provides the formal equations for the physics-guided descriptors described in Section 2.2.

A.1 Goodenough-Kanamori Superexchange

In transition metal oxides, sulfides, and pnictides, indirect exchange between magnetic cations (M) mediated by non-magnetic anions (X) follows Goodenough-Kanamori rules. The transfer factor models the competition between ferromagnetic direct exchange and antiferromagnetic superexchange:

$$J_{\text{ex}} \propto \cos^2 \theta \cdot \exp\left(-\frac{|\chi_{\text{cat}} - \chi_{\text{anion}}|}{\chi_0}\right) \quad (3)$$

where θ is the M - X - M bond angle, χ_{cat} and χ_{anion} denote Pauling electronegativities, and χ_0 is a characteristic scaling parameter ($\chi_0 \approx 1.5$). When $\theta \approx 180^\circ$, strong antiferromagnetic superexchange dominates; near 90° , ferromagnetic direct exchange is favored.

A.2 Dzyaloshinskii-Moriya Interaction (DMI) Proxy

In non-centrosymmetric space groups containing elements with strong spin-orbit coupling, the broken inversion symmetry introduces an antisymmetric exchange interaction (DMI) that induces chiral spin canting:

$$D_{\text{proxy}} = \mathbb{I}(\text{non-centrosymmetric}) \sum_i x_i \left(\frac{Z_i}{100}\right)^4 \cdot \mathbb{I}(Z_i \geq 40) \quad (4)$$

where $\mathbb{I}(\cdot)$ is an indicator function (equal to 1 if the condition holds, 0 otherwise), x_i is the atomic fraction of element i , and Z_i is its atomic number. The fourth-power dependence reflects the relativistic scaling of spin-orbit coupling ($\lambda_{\text{SOC}} \propto Z^4$).

A.3 Crystal Field and Stevens Operator Hamiltonian

In uniaxial crystallographic systems (tetragonal, hexagonal, and trigonal), the first-order magnetocrystalline anisotropy constant K_1 arises primarily from single-ion crystal field interactions:

$$\mathcal{H}_{\text{CF}} = B_2^0 O_2^0 + B_4^0 O_4^0 \quad (5)$$

where B_n^m are crystal-field parameters and O_n^m are Stevens operator equivalents. The second-order operator is defined by:

$$O_2^0 = 3J_z^2 - J(J+1) \quad (6)$$

where J is the total angular momentum quantum number. The sign of the second-order Stevens factor α_J dictates whether the single-ion interaction favors an easy axis ($\alpha_J < 0$, uniaxial anisotropy $K_1 > 0$) or an easy plane ($\alpha_J > 0$).

B MagFormer Architecture Details

This appendix provides the network equations for the MagFormer composition graph transformer described in Section 2.2.

B.1 Node Initialization and Projection

Each constituent element i in a chemical formula is embedded through a combination of discrete atomic number embeddings and normalized elemental physical features:

$$\mathbf{h}_i^{(0)} = \mathbf{W}_{\text{node}} [\mathbf{e}_{Z_i} \parallel x_i \parallel \mathbf{p}_i] \quad (7)$$

where $\mathbf{e}_{Z_i} \in \mathbb{R}^{64}$ is a learned element embedding vector, $x_i \in (0, 1]$ is the atomic fraction, and $\mathbf{p}_i$ is a normalized vector of elemental physical properties (electronegativity, covalent radius, valence, and Stoner parameter).

B.2 Pair-Bias Self-Attention

Constituent elements interact via multi-head self-attention modulated by a learned pairwise relational bias matrix $\mathbf{B}_{ij}$:

$$\mathbf{A}_{ij} = \text{softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_j^T}{\sqrt{d_k}} + \mathbf{B}_{ij} \right) \mathbf{V}_j \quad (8)$$

where query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) projections are computed from $\mathbf{h}_i^{(l)}$. The pairwise bias $\mathbf{B}_{ij}$ is generated by passing elemental feature differences $|\mathbf{p}_i - \mathbf{p}_j|$ through a two-layer multi-layer perceptron.

B.3 Leakage-Free Composition Pooling

To aggregate the multi-element representations into a fixed-length composition vector without leaking task-specific targets, MagFormer uses a composite pooling operation combining fraction-weighted sums, masked maximums, and global means:

$$\mathbf{z}_{\text{comp}} = \left[\sum_{i=1}^N x_i \mathbf{h}_i^{(L)} \parallel \max_{i=1 \dots N} \mathbf{h}_i^{(L)} \parallel \frac{1}{N} \sum_{i=1}^N \mathbf{h}_i^{(L)} \right] \in \mathbb{R}^{384} \quad (9)$$

where L is the final transformer layer and N is the number of elements in the formula. The resulting 384-dimensional representation encodes complex non-linear chemical interactions and is supplied as an input to the tabular ensemble models.